

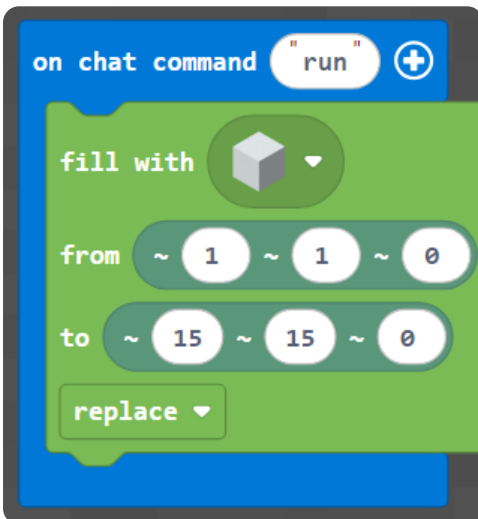
3D Coordinates · Pixel Art on the Wall (x, y) — English Worksheet (Advanced)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

This lesson you copy a picture made of blocks onto your wall. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict

Read each set of steps. For each one, write **what you think you will see** and **why**.

```
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

Across or up? Which number stays the same, and why does that decide the direction?


```
place green block at (8, 13)
place green block at (8, 14)
place green block at (8, 15)
```

Near the top or the bottom? Across or up? Explain how you know.


```
place yellow block at (6, 1)
place yellow block at (7, 2)
place yellow block at (8, 3)
```

Both numbers grow by 1 each time. Row, column, or a stair going up? Why?


```
place yellow block at (5, 1)
place yellow block at (6, 1)
place yellow block at (5, 2)
place yellow block at (6, 2)
```

Four blocks together. What shape do they make, and where does it sit? Explain your reason.

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — These should make the **bottom row of the pineapple**: five yellow blocks from (6, 1) to (10, 1). Right now they all land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The apple stem should go **up** at $x = 8$, at (8, 13), (8, 14), (8, 15). Right now it grows **across** instead.

```
place brown block at (13, 8)
place brown block at (14, 8)
place brown block at (15, 8)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran code that matched the grid exactly. Every block landed at an x **one bigger** than the grid says, so the whole picture shifted across — even though the code is fine.

```
(the code matched the grid exactly)
```

What did your friend forget to do before pressing run? Why does the home spot change where (x, y) lands, even when the code is correct?

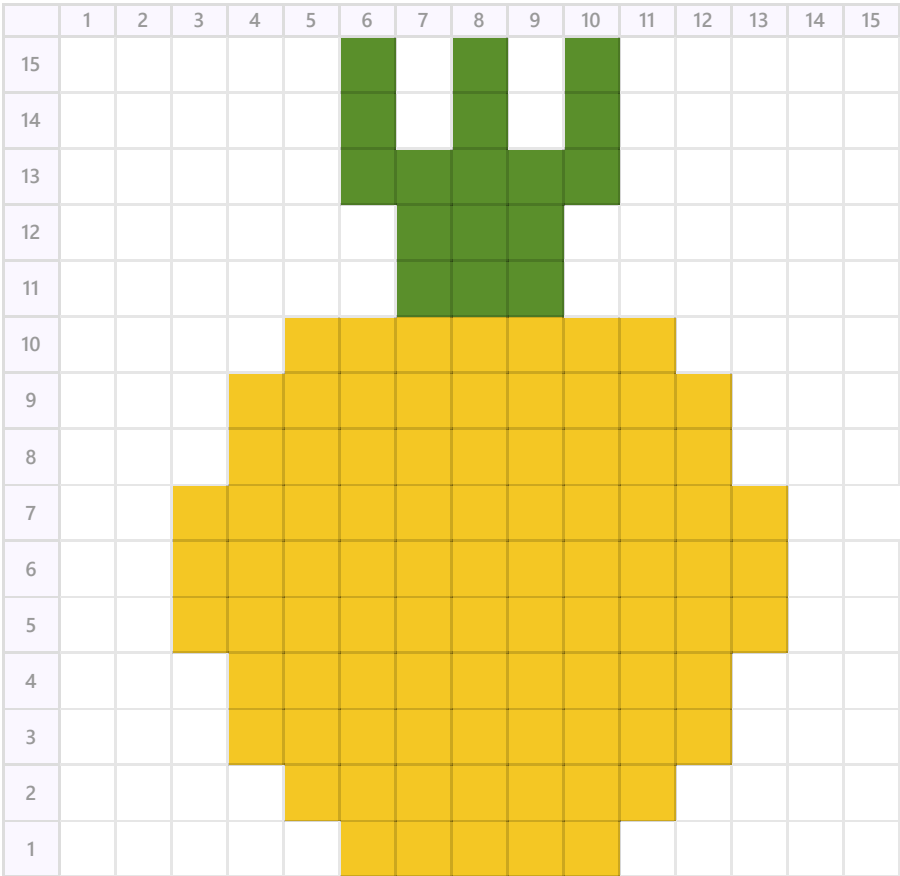
3 · Show Your Work 📷 🎥

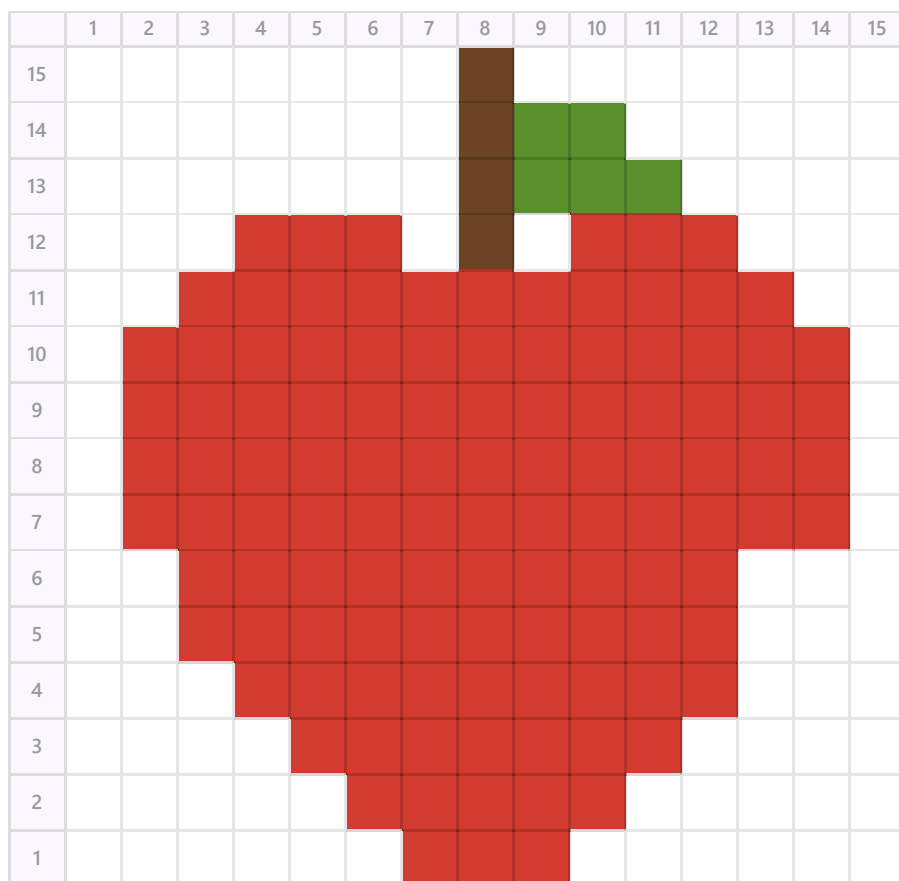
Switch to your homework world and stand on your **home spot** (the red block). The numbers across the **top** are **x**, the numbers down the **side** are **y**.

Plan first. Before you place anything, look at the 🍍 pineapple and write the (x, y) of its **bottom-left** block and its **top** block. This tells you how wide and how tall it is.

Copy a picture below onto your wall, block by block, by reading the (x, y) of every colored square.

🍍 Pineapple





Modify challenge. After you finish one fruit, change something of your own — swap a colour, add a leaf, or make the picture one block bigger. Write the (x, y) of the blocks you changed or added.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.